

Ali Vaziri

Machine Learning for Autonomous Systems

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I'm doing my Ph.D. research in machine learning, optimization, and optimal control for autonomous systems. My Ph.D. research spans ML-based dynamical modeling, Bayesian inference, and GPU-accelerated model predictive inferential control.

I worked at Cruise (General Motors) and Mitsubishi Electric Research Laboratories (MERL). At Cruise, I built terabyte-scale data pipelines and multi-GPU training pipelines for scalable scene reconstruction; at MERL, I developed physics-constrained ML models for thermofluid systems and worked on predictive control of HVAC systems.

Industry Experience

Cruise – General Motors

AI/ML Software Engineering Intern

Sunnyvale, CA | Summer 2026

ML-Based Vehicle Motion Reconstruction

- Built an ML system to reconstruct missing vehicle motion in recorded-driving simulations; reduced best-candidate average position error by **24%** versus the team's prior baseline on approximately **58,000 held-out vehicle trajectories**.
- Built large-scale data and featurization pipelines for **38,000+ highway scenes** (scale in TBs of raw tracking data) for ML training.
- Developed a training pipeline at scale using **multiple H100 GPUs**, reducing ML model training time from **12 to 3 hours**.

Mitsubishi Electric Research Laboratories (MERL)

ML and Optimization Research Intern

Cambridge, MA | Summer 2024

- Developed a physics-constrained Neural ODE–GRU model for vapor-compression HVAC systems, improving forecasting accuracy by **14%** and achieving a **5.7× runtime speedup** versus the prior modeling approach.
- Delivered reusable PyTorch software for data processing, training, evaluation, and GPU-accelerated control experiments with limited real-world data.

Research Experience

Graduate Research Assistant

University of Kansas (2023–2025); Michigan State University (2026–present)

2023–present

- Developed inferential model predictive control for learned vehicle models, high-dimensional fluids, and thermofluid systems; tested configurations ran up to **200× faster** than IPOPT on GPU in tested benchmarks.
- Developed GPU-compatible tensor-variate ensemble Kalman smoothing for spatial models, with up to **18.6× faster computation** and approximately **11× lower memory use** than a vectorized GPU baseline.
- Designed motion planning with Student's- t distributions, sequential Monte Carlo, and ensemble Kalman smoothing for non-Gaussian uncertainty and outliers; trained GRU and ResNet vehicle models on real driving data for Bayesian planning and trajectory optimization.
- Developed continuous-time optimal control of Neural ODEs for soft-robot manipulators.

Education

Michigan State University — Ph.D. in Mechanical Engineering

Expected graduation: December 2027. Prior Ph.D. study: University of Kansas, 2023–2025 (transferred).

2026–present

University of Kansas — M.S. in Mechanical Engineering; GPA: 4.00/4.00

2023–2025

Sharif University of Technology — B.S. in Marine Engineering

2016–2021

Technical Skills

Programming and ML: Python, C/C++, PyTorch, TensorFlow; VAEs, diffusion models, LLMs, large-scale training pipelines.

GPU and data systems: DDP, NCCL, mixed precision, Google Cloud Storage, Dataflow, large-scale data pipelines.

Control and simulation: MPC, Bayesian inference, Kalman filtering/smoothing, particle filtering, trajectory optimization; ROS, MuJoCo, PyBullet, CasADi/IPOPT, MATLAB/Simulink.

Development: Linux, Git, Bazel, performance profiling, unit and integration testing.

Publications

I. Askari, **A. Vaziri**, et al. “Model Predictive Inferential Control of Neural State-Space Models for Autonomous Vehicle Motion Planning.” *IEEE Transactions on Robotics*, 2025.

A. Vaziri et al. “Modular Deep Learning Framework for Vapor-Compression HVAC Systems: Scalable, Efficient, and Physically Consistent Modeling.” *Applied Thermal Engineering*, 2026.

A. Vaziri et al. “Bayesian Inferential Motion Planning Using Heavy-Tailed Distributions.” *ACC*, 2025.

A. Vaziri et al. “Continuous-Time Optimal Control of Neural ODEs via Bayesian Inference.” *IEEE CDC*, 2026.

A. Vaziri et al. “Optimal Inferential Control of Convolutional Neural Networks.” *ACC*, 2025.